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1. $t = 500 \text{ days}$ $T = 625 \text{ K}$ $\delta^0 = 600 \mu\text{m}$

a) $\delta(\mu\text{m})$ $t^* = 6.62 \times 10^7 \exp\left(\frac{11949}{T}\right) \rightarrow t^* = 133.007 \text{ days}$, $t > t^*$

$\delta^* = 5.1 \exp\left(\frac{-550}{T}\right) \rightarrow \delta^* = 5.1 \exp\left(\frac{-550}{625}\right) \rightarrow \delta^* = 2.115 \mu\text{m}$

$K_L\left(\frac{\mu\text{m}}{\text{d}}\right) = 7.48 \times 10^6 \exp\left(\frac{-12500}{T}\right) \rightarrow K_L = 0.01542$

$\delta(\mu\text{m}) = \delta^* + K_L(t - t^*) \rightarrow \delta(\mu\text{m}) = 2.115 + 0.01542(500 - 133.007)$

$\delta(\mu\text{m}) = 7.774 + \delta^0 = 607.774 \mu\text{m}$ δ at $t = 500 \text{ days}$

- only oxide thickness

b) $C_H^{\text{clad}} (\text{wt. ppm})$? $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$, $f = 14\%$, $f_{\text{ZrO}_2}^0 = \frac{15.99 \cdot 2}{15.99 \cdot 2 + 91.22} = 0.2596$

$\frac{M_H}{M_o} = \frac{1}{16}$

$\delta = 607.77 \mu\text{m} = 0.06077 \text{ cm}$

$C_H = \left(\frac{2f \cdot \delta - \rho_{\text{ox}} \cdot f_{\text{ZrO}_2}^0 \cdot \frac{M_H}{M_o}}{\left(t - \frac{\delta}{\text{PBR}}\right) - \rho_{\text{metal}}} \right) \cdot 10^6 \rightarrow C_H = \left(\frac{2(14)(0.06077 \text{ cm})(5.68)(0.2596)\left(\frac{1}{16}\right)}{\left(500 - \left(\frac{0.06077}{1.56}\right)\right) \cdot 6.5} \right) \cdot 10^6$

oxide
not time
600 μm

$C_H = 0.483 \text{ wt. ppm}$

2. Diffusion of oxygen is the rate limiting step. Cannot have corrosion without O_2 ; O movement is essential
3. Hydrides form at solubility and Soret effect areas. Outside of cladding. Soret effect: H moves to low temp region. Solubility is higher on crack tip^{due to stress}. Cracks on outside of cladding; Delayed Hydride cracking. H moves to crack tip. Hydrides form circumferentially along highest hoop stress. Higher hoop stress is increased solubility. Can form blisters. Hydrides embrittle Zr cladding. Hydrides form path of increased probability of crack propagation. Can form radially in fact.

4. RIA: Reactivity induced Accident. i.e. accident caused by sudden, sharp increase in reactivity.

PWR RIA: control rod ejection. milliseconds

BWR RIA: blade drop. Minutes

Events: Reactivity spike causes higher fuel temp, increased swelling $\rightarrow$ leads to Departure from nucleate boiling or PCMI. DNB causes steam, temperature of fuel continues to rise $\rightarrow$ can have fuel melting and/or cladding plastic deformation. If cladding cracks, can have fuel ejection. Fuel ejection can lead to local power peaks.

If PCMI, FP_s increase pressure inside tube, can lead to cracking and fuel ejection.

- Burnup plays major role in how system reacts to RIA. More burnup means more/thicker oxide layer $\rightarrow$ less ductile cladding $\rightarrow$ PCMI. Higher burnup fuel $\rightarrow$ higher T centerline $\rightarrow$ higher possibility of T melt because less heat transfer from fuel to steam.

5. LOCA: loss of coolant accident. Accident caused by loss of coolant or lack of flow.

A LOCA can lead to similar results as a RIA but is caused differently.

If coolant is lost through burst pipe, then pressure of coolant drops $\rightarrow$ increased risk of PCMI/Ballooning of cladding. Also, leads to DNB which causes less heat transfer from fuel to coolant.

coolant flow can be impeded by plastic deformation of cladding tubes $\rightarrow$ no coolant channels $\Rightarrow$ increased fuel temp. High plastic deformation of cladding $\rightarrow$ rupture of cladding $\Rightarrow$ ejection of fuel. Still function of Burnup similar to RIA.

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6. Burnup of fuel also means increased FP's present and more oxidation layer on cladding = more brittle. ~~Burnup~~ Increasing burnup leads to increasing failure probability. There are many criteria to limit this. Limit on dissolved Hydrogen (cladding embrittlement), Limit on FR amount. And others. To ~~control~~ keep system below limits of increased failure probability.

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7. Accident tolerance pathways: 1. Improved cladding properties (thermodynamic and mechanical)
2. Improved fuel properties (higher T_{melt} / ~~dopants for slower diffusion~~ better thermal cond.)
3. Improved steam corrosion resistance (Zn addition, pH balance)
4. Fission product retention (dopants in fuel $\rightarrow$ increased grain size $\rightarrow$ decreased FP mobility)

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8. Near-term ATF = cladding coating. MAX: Cr-Al/Si-C/N. Oxide and Metallic coating options. Coating cladding limits ZrO_2 formation, maintains cladding ductility. Oxide coatings are more resistant to corrosion, but more CTE incompatible with Zr cladd. Metallic coatings not as strong at resisting corrosion.

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9. High temp steam causes high corrosion to Zr cladding. $Zr + 2H_2O \rightarrow ZrO_2 + H_2$. Forms oxide layer, ZrO_2 , and H_2 gas enters cladding. Oxide layer grows and becomes passivating - protective of the Zr cladding. PBR ratios between 1 and 2, ratio of oxide to metal. If oxide layer exceeds PBR of 2, it can flake off and expose Zr cladding. The H_2 embrittles the Zr cladding by forming hydrides. Exposed cladding, after oxide flake, embrittled by hydrides, can crack cladding. Crack propagation, enhanced by delayed hydride cracking = hydride solubility increase on crack tip.
- almost all

10. Limiting phenomena: 1. Burnup below $\sim 5\%$. Burnup increases accident probability.
2. Cladding oxide layer — total O/M ratio. ~~then~~ increased oxide thickness = probability of flaking and crack propagation.
3. Plastic deformation of cladding. PCMI, pellet mechanical stress on cladding, increases hoop stress $\rightarrow$ increases hydride $\rightarrow$ embrittlement of cladding.
- general concepts, but not what I was looking for

11. CRUD: fuel performance impacts: lower heat transfer to coolant $\rightarrow$ higher T centerline, also can have Boron, so decreased neutron economy $\rightarrow$ CIPS, power shifts due to this.
- Safety impacts: CRUD becomes irradiated, can transition to soluble metals in coolant, redeposit outside of reactor core. Exposure to radiation on heat exchanger.

12. pH control and Zn addition. pH limits by adding Boric acid and LiOH. pH > 7.1 leads to decreased CRUD formation. Also, decreased oxidation.
- Zn addition - Zn forms more stable spinel structure of CRUD. It also has lower formation energy than common CRUD metals like Cr/Cu/Ni so it takes the position in CRUD, not allowing these metals to become irradiated.

13. MOX microstructure phenomena: 1. Grain size change. Void in center, columnar grains, equi-axed grains, as-sintered. Columnar grains form above 1800°C . Voids move to higher temp axially and redeposit U behind. Void movement to center of fuel creates central void. Temp of fuel can be measured by extent of columnar grains.

2. Metal FP precipitates form more readily in equi-axed region. FPs can move to outside of fuel and create JOG layer which causes PCCl with Stainless Steel cladding.